

Assignment 02

University of Sialkot

Department of Software Engineering

Faculty of Software Engineering

Web Engineering

Guiding Principles: Smart Diet Planner(TandooriFit)

Submitted To:

Sir Museb Khalid

Dated: 30/06/2026

By:

Muhammad Jasim 23010101-054

Atiqah Asif 23010101-016

Introduction

Smart Diet Planner is a client-side web application designed to help individuals plan and manage a healthy diet in a simple and personalized way. It allows users to calculate essential health metrics such as BMI, BMR, TDEE, and daily calorie requirements based on their age, gender, body measurements, activity level, and fitness goals. Based on these results, the application assists users in creating a structured daily meal plan with macro-nutrient targets and flexible meal distribution, making diet planning more practical and user-friendly.

The application is useful for users who want a clear and guided approach to nutrition without complex manual calculations. Many people find it difficult to estimate calories and macros or to design balanced meals that align with their fitness goals. Smart Diet Planner addresses this problem by providing automated calculations, customizable meal planning, and a flexible food database that can be adapted or extended in future versions. By focusing on simplicity, clarity, and modular design, the application lays a strong foundation for future enhancements such as online data storage, user accounts, and dynamic food databases.

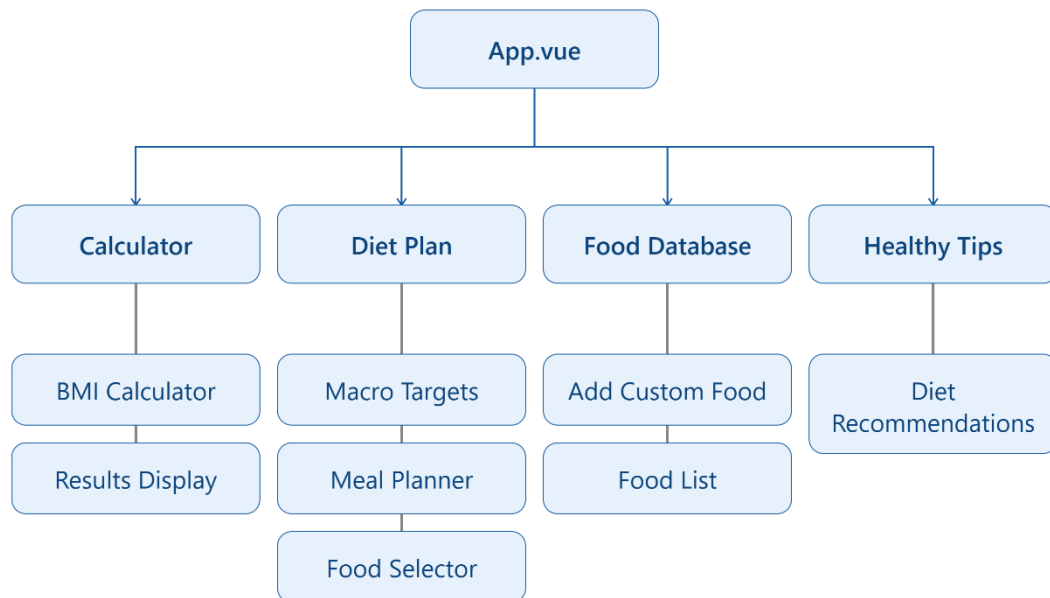
Project Features

- **BMI Calculator**
Calculates Body Mass Index using user-provided height and weight and categorizes the result to help users understand their current health status.
- **Daily Calorie Calculator**
Computes daily calorie requirements based on BMR, activity level, and fitness goals such as weight maintenance, loss, or gain.
- **Meal Planner**
Generates a structured daily meal plan according to calorie targets, selected diet style, and number of meals per day.
- **Macro-Nutrient Breakdown**
Provides detailed information about daily protein, carbohydrate, and fat requirements to support balanced nutrition.
- **Flexible Food Database**
Allows users to manage food items and nutritional values, with support for modifying or extending the food list as needed.
- **Diet Recommendations**
Offers healthy dietary tips and guidance to promote better eating habits and informed food choices.
- **User-Friendly Dashboard**
Presents calculated results and meal plans in a clear and organized interface for easy monitoring.

- **Responsive Single Page Application**

Ensures smooth navigation and usability across different screen sizes using client-side routing.

Architecture Tree:



Vue Router

- **/ → CalculatorView**
(BMI, BMR, TDEE, and daily calorie calculator)
- **/plan → PlanView**
(Daily meal planner with macro breakdown and meal management)
- **/database → DatabaseView**
(Food database management with add/edit and macro estimation)
- **/tips → TipsView**
(Healthy diet tips and nutrition guidance)

Components

- **App.vue** → Root component that manages layout, navigation, and shared application state
- **BmiCalculator.vue** → Takes user input for BMI, BMR, TDEE, and calorie calculations
- **BmiResults.vue** → Displays calculated health results such as BMI category and daily calorie needs
- **CalculatorView.vue** → Combines calculator input and results into one page

- **PlanView.vue** → Handles meal planning, meal updates, and macro tracking
- **DatabaseView.vue** → Manages food database and custom food entries
- **TipsView.vue** → Displays healthy diet tips and nutrition guidance
- **macroEstimator.js (utility)** → Automatically estimates food macros based on name keywords
- **router/index.js** → Manages navigation between different views in the SPA

Props and Emits

In Vue.js, **props** and **emits** are used for communication between components.

Props are used to send data from a parent component to a child component. This allows child components to display or use data provided by the parent.

Emits are used when a child component wants to send information or trigger an action in the parent component. It works like sending a message upward.

Example code of BmiCalculator Component including props and emitters:

```
const props = defineProps({
  units: String,
  age: Number,
  gender: String,
  weightKg: Number,
  weightLbs: Number,
  heightCm: Number,
  feet: Number, inches:
  Number,
  activity: String,
  goal: String,
  planStyle: String,
  mealsPerDay: Number
});

const emit = defineEmits([
  'update:units', 'update:age', 'update:gender', 'update:weightKg',
  'update:weightLbs', 'update:heightCm', 'update:feet', 'update:inches',
```

```
'update:activity', 'update:goal', 'update:planStyle', 'update:mealsPerDay',  
'calculate', 'reset'  
]);
```

Screenshots:

Calculator:

The screenshot shows the TandooriFit calculator interface. At the top, the logo 'TandooriFit' is displayed in green, with a subtitle 'Estimate your BMI, BMR, TDEE, and daily calorie target with local Pakistani foods.' To the right of the logo are two buttons: 'Switch to Imperial' and 'Dark'. Below the header is a navigation bar with four tabs: 'Calculator' (highlighted in green), 'Diet Plan', 'Food DB', and 'Healthy Tips'. The main form contains several input fields: 'Age' (28), 'Gender' (Male), 'Weight (kg)' (70), 'Height (cm)' (175), 'Activity' (Moderate), 'Goal' (Maintain), 'Plan Style' (Balanced), and 'Meals per Day' (3 meals). Below these fields are two buttons: 'Calculate' (green) and 'Reset' (grey). The results section, titled 'Your result', shows a 'HEALTHY' status. It displays four key metrics: BMI (22.9), IDEAL WEIGHT (56.7 - 76.3 kg), BMR (1659 kcal/day), and TDEE (2571 kcal/day). At the bottom, a bar chart shows 'MAINTENANCE CALORIES' with a value of 2571 kcal/day.

Metric	Value
BMI	22.9
IDEAL WEIGHT	56.7 - 76.3 kg
BMR	1659 kcal/day
TDEE	2571 kcal/day
MAINTENANCE CALORIES	2571 kcal/day

Diet Plan:

TandooriFit

Estimate your BMI, BMR, TDEE, and daily calorie target with local Pakistani foods.

Switch to Imperial

Dark

Calculator

Diet Plan

Food DB

Healthy Tips

Custom Diet Plan

Balanced meals with steady energy

CALORIES

2571 kcal

PROTEIN

161g

CARBS

289g

FAT

86g

Track what you ate

CHOOSE FOOD

Select a food item

CALORIES

250

PROTEIN (G)

20

CARBS (G)

30

FAT (G)

10

Add to Meal Plan

Active Daily Meals

Breakfast

No food selected | 720 kcal | P: 45g | C: 81g | F: 24g

UPDATE FOOD CHOICE

Choose a database food

Swap Suggestion

Remove

Lunch

No food selected | 900 kcal | P: 56g | C: 101g | F: 30g

UPDATE FOOD CHOICE

Choose a database food

Swap Suggestion

Remove

Food Database:

TandooriFit

Estimate your BMI, BMR, TDEE, and daily calorie target with local Pakistani foods.

Switch to Imperial

Dark

Calculator

Diet Plan

Food DB

Healthy Tips

Food Database Manager

Create or manage local food options available in your diet selectors.

Add Food to Database

FOOD ITEM NAME

e.g. Paratha, Chicken Tikka

☒

Auto-calculate macros based on name lookup

CALORIES

250

PROTEIN (G)

20

CARBS (G)

30

FAT (G)

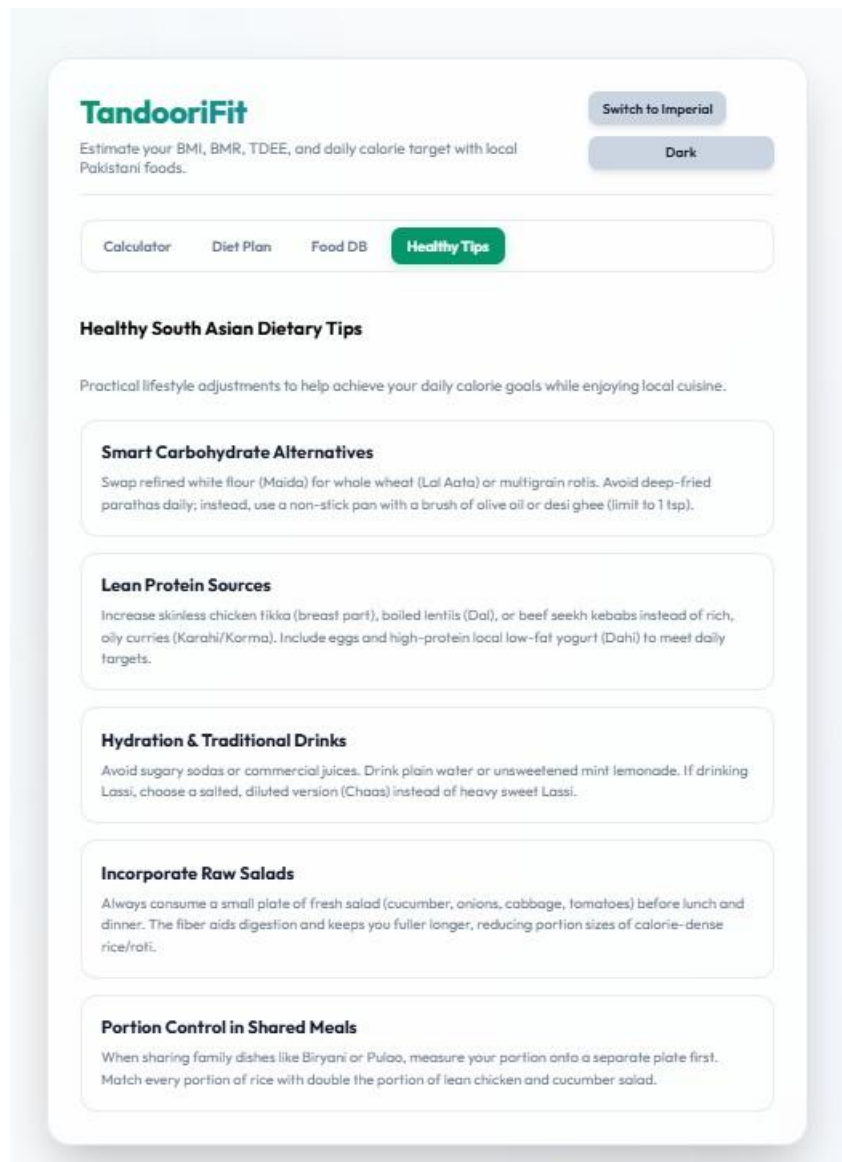
10

Save Food to Database

Registered Foods (16)

NAME	CALORIES	PROTEIN	CARBS	FAT
Roti (Whole Wheat Chapati)	120 kcal	3.5g	24g	0.5g
Basmati Rice (Boiled, 1 cup)	205 kcal	4.2g	44.5g	0.4g
Chicken Biryani (1 plate)	450 kcal	22g	58g	12g
Dal (Lentil Curry, 1 cup)	180 kcal	9.5g	28g	4g
Chicken Karahi (150g)	310 kcal	26g	5g	20g
Mixed Sabzi Curry (1 cup)	150 kcal	3g	16g	8g
Chana Chaat (1 cup)	220 kcal	7.5g	36g	4.5g

Healthy Tips:



Github Repository Link:

<https://github.com/jasimjawediqbal/Smart-Diet-Planner>

Conclusion

The Smart Diet Planner frontend was developed using Vue 3 and Vite, focusing on a modern and modular single-page application architecture. The system uses Vue Router for smooth navigation between different views, while reusable components ensure a clean and organized code structure. Communication between components is handled using props and emits, following proper parent-child data flow principles. This frontend serves as a strong foundation for the project, and the backend integration will be developed in the next phase to extend its functionality.